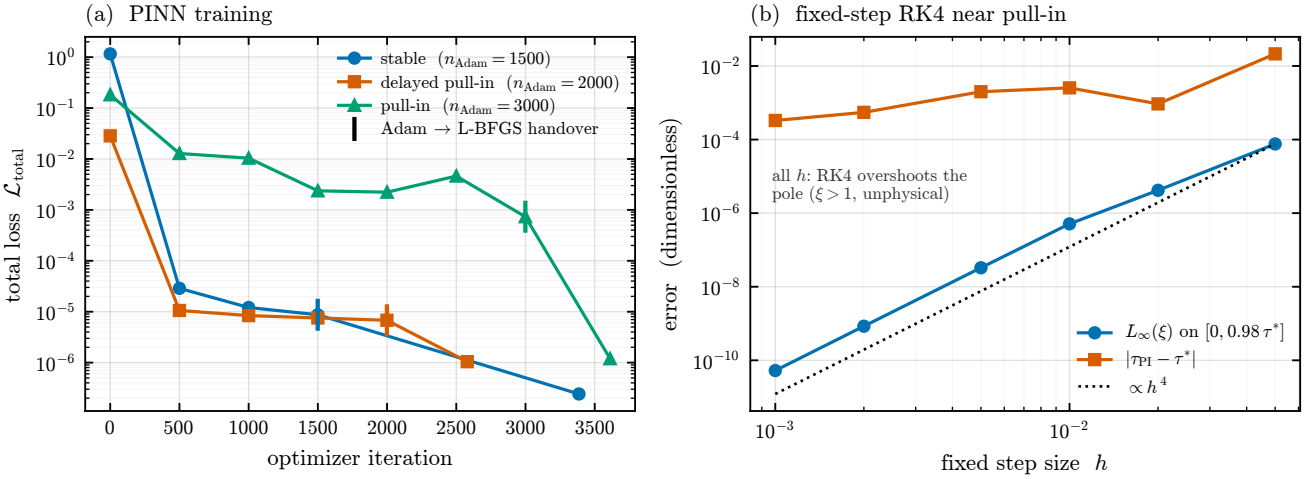
(a) PINN training
stable ($n_{\text{Adam}} = 1500$)
delayed pull-in ($n_{\text{Adam}} = 2000$)
pull-in ($n_{\text{Adam}} = 3000$)
Adam → L-BFGS handover
total loss $\mathcal{L}_{\text{total}}$
optimizer iteration
(b) fixed-step RK4 near pull-in
all h: RK4 overshoots the pole ($\xi > 1$, unphysical)
$L_\infty(\xi)$ on $[0, 0.98\,\tau^*]$
$|\tau_{\text{PI}} - \tau^*|$
$\propto h^4$
error (dimensionless)
fixed step size h